

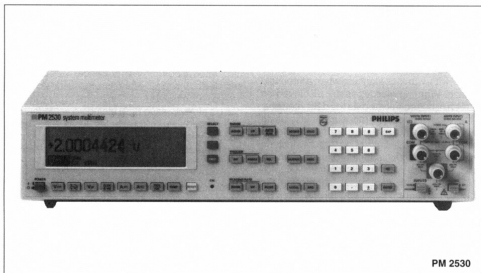
Philips PM 2530 -- Fluke and Philips Test and Measurement Catalog 1992

Bench/System Multimeters

PM 2530



NEW



PM 2530

PM2530

20,000,000 counts
90 day accuracy to 18 ppm
20 ohm range with 10 uohm resolution
V/f function for a.c. Volts 2 mHz - 500 Hz
1000 r/s via IEEE-bus or 50,000 r/s in burst mode
Temperature, frequency and period measurement

For systems and bench

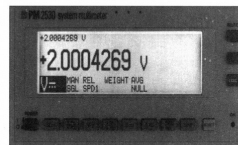
The PM2530 provides both the high degree of accuracy and fast processing speeds needed in manufacturing and R&D environments. As a result, throughput time can be reduced in a variety of applications. Time saving is also possible in programming of the instrument, thanks to logical command structures that comply with IEEE-488.2 and SCPI standards (SCPI, Standard Commands for Programmable Instruments). All control parameters are easily accessed either from the front panel or via the IEEE interface, allowing quick and precise set-up.

Research into the use of multimeters has led to a design that is at home both on the bench and in a system. For this reason, the PM2530 fits into a standard 19" rack. In addition, the unit has

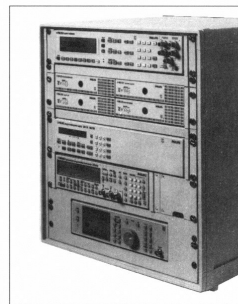
switchable front/rear inputs, giving the operator maximum flexibility for checking and testing, especially when in a system configuration. For bench use, the sturdy construction and ergonomic design guarantee optimum performance.

Easy to learn

Another important aspect of the PM2530 is its user interface, which combines a logical layout of front panel controls with menu-driven command screens. Together, these reduce the need to rely on the instrument's handbook by guiding operators through operating procedures. This minimises the time needed for instrument familiarisation, and enhances operation for practised users. Furthermore, whenever processed values are displayed, original measured values are also visible as a convenient aid to the operator.

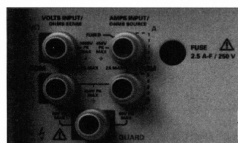


Control keys are arranged in groups that relate to the type of functions required; for example primary measurement functions are located together, and are directly accessible; more sophisticated functions are available through combinations of keys and drop-down menus. Results are easy to read thanks to the large characters on the backlit LCD, and current instrument status is clearly displayed by function labels that appear at the bottom of the screen. Fast automatic ranging is also available, enabling the operator to begin taking measurements without the need for set up procedures.

Built for systems

Besides meeting the requirements of the IEEE-488.2, the PM2530 provides the programming convenience of SCPI. This has become an accepted standard for most instrument manufacturers, and makes it simpler to configure or modify the system when other combinations of instruments are required.

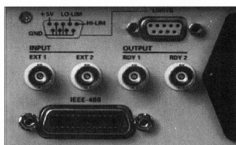
Programming possibilities are extensive. For example, besides interval and timing programs, the PM2530 has hardware trigger and ready lines. These permit better synchronisation with signal sources, scanners and other instruments.



This page shows features and specifications of the Philips PM 2530.

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Furthermore, there is a proprietary feature called "Process Programming", which allows operators to enter several commands in advance as time becomes available in the bus communications. These various commands can then be triggered by hardware signals, yielding faster execution of commands when time is unavailable in the test cycle. In turn, this means better synchronisation with the outside world, and higher throughput in repetitive measurement applications.

Convenient. Powerful

The PM 2530 incorporates many features that increase its applications possibilities. For example, minimum and maximum values since last function selection can be stored if required. Limits are definable, with warnings triggered when a threshold is crossed. Warning messages are displayed on the screen or communicated via the data interface, and the total number of measurements falling above, within and below the threshold is memorised. Powerful processing features include on-board averaging to reduce processing time in external computer systems and minimises "bus load". In addition, it is possible to program the number of measurements used for the average value, or a weighted average according to the following formula:

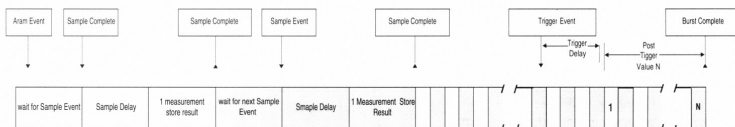
$$\langle X \rangle = \frac{W-1}{n} \langle X \rangle + \frac{1}{n} X_n$$

Where X is the measured value and W is programmable between 1 and 100. Programmable scale factors allow correct

quantification of transducer values, and analyses are possible on differences in measured values, with results displayed as a percentage of a measured or programmed value. If required, measured values are converted to dBm, and calculated according to a reference measurement selected by the relative function. For fast instrument set-up, 10 complete sets of instrument parameters can be stored, (save 0..9) including scale factors and relative values, thresholds, limits, etc. These files are ready for instant recall at the push of a button. At power-up the instrument will revert to the default setting, which is stored via "save o".

Simple calibration and convenient diagnostics

Through the IEEE-488.2 interface or front panel it is possible to calibrate the instrument electronically, without removing the cover. Not only does this save time but costs as well, while making it easy to ensure that the meter delivers the best possible results. The PM2530 also has the convenience of auto-diagnostics, and when required can perform a self-test routine.



High accuracy and speed

The PM2530 has a basic accuracy of 18 ppm over 90 days, with a maximum display length of 20,000,000 counts providing a resolution of 100 nV. For resistance measurements, a measurement with a resolution of 10 µΩ is possible. Offset voltage and thermal voltage can be compensated for by "nulling" in all ranges.

For high throughput, the PM2530 can take and transfer up to 1,000 measurements per second to the interface, with an resolution of 10 µV. For greater measurement speed, the internal buffer can store 15,000 measurements. This facilitates a "burst mode" of 50,000 measurements per second, at a resolution of 100 µV. Where 10 µV resolution is still required, 4,000 measurements per second can be taken and stored. The special "burst mode" can be started by an external trigger signal or via commands received manually or via an interface command. The number of measurements after a second trigger can also be programmed, permitting both pre- and post-trigger measurements to be made.

Specifications

DCVOLTAGE.

General

Range	Full scale	Maximum resolution	Temperature coefficient ¹ (ppm reading + ppm range)/K	Input impedance
20 mV	200 000	100 nV	±(10 + 15)	1 GΩ // 150 pF
200 mV	2 000 000	100 nV	±(10 + 15)	1 GΩ // 150 pF
2 V	20 000 000	100 nV	±(5 + 0.15)	1 GΩ // 150 pF
20 V	20 000 000	1 µV	±(5 + 0.15)	1 GΩ // 150 pF
200 V	20 000 000	10 µV	±(10 + 0.15)	10 MΩ // 150 pF
2000 V ²	20 000 000	100 µV	±(10 + 0.15)	10 MΩ // 150 pF

Accuracy³ (ppm of Reading + ppm of Range)

Range	Speed 1			Range	Speed 2		
	24 Hour	90 Day	1 Year		24 Hour	90 Day	1 Year
20 mV	36 + 10	50 + 20	60 + 20	20 mV	36 + 10	50 + 20	60 + 20
200 mV	18 + 3	25 + 4	36 + 4	200 mV	18 + 3	25 + 4	36 + 4
2 V	15 + 2	18 + 2	24 + 2	2 V	15 + 2	18 + 2	24 + 2
20 V	15 + 2	18 + 2	24 + 2	20 V	15 + 2	18 + 2	24 + 2
200 V	18 + 2	25 + 2	36 + 2	200 V	18 + 2	25 + 2	36 + 2
2000 V ²	18 + 2	25 + 2	36 + 2	2000 V ²	18 + 2	25 + 2	36 + 2

¹ Additional error from $T_{ref} \pm 5^\circ C$

² Maximum input voltage 1000V

³ Specifications relative to calibration values. At factory delivery add 4.5 ppm to ppm of reading for absolute accuracy at 10V.

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Range	Speed 3			Speed 4		
	24 Hour	90 Day	1 Year	24 Hour	90 Day	1 Year
20 mV	40 + 10	50 + 20	60 + 20	150 + 100	200 + 100	250 + 50
200 mV	20 + 10	25 + 10	35 + 10	100 + 50	150 + 50	200 + 50
2 V	15 + 5	20 + 5	25 + 5	50 + 50	100 + 50	150 + 50
20 V	15 + 5	20 + 5	25 + 5	50 + 50	100 + 50	150 + 50
200 V	20 + 5	20 + 5	35 + 5	50 + 50	100 + 50	150 + 50
2000 V ²	20 + 5	20 + 5	35 + 5	50 + 50	100 + 50	150 + 50

Range	Speed 5			Speed 6		
	24 Hour	90 Day	1 Year	24 Hour	90 Day	1 Year
20 mV	100 + 100	150 + 100	200 + 100	500 + 500	1000 + 1000	1000 + 1600
200 mV	100 + 100	150 + 100	200 + 100	500 + 500	1000 + 1000	1000 + 1000
2 V	100 + 100	150 + 100	200 + 100	500 + 500	1000 + 1000	1000 + 1000
20 V	100 + 100	150 + 100	200 + 100	500 + 500	1000 + 1000	1000 + 1000
200 V	100 + 100	150 + 100	200 + 100	500 + 500	1000 + 1000	1000 + 1000
2000 V ²	100 + 100	150 + 100	200 + 100	500 + 500	1000 + 1000	1000 + 1000

Noiserejection(dB)

Speed	Max. scale length	NMR Linefreq.	NMR 400 Hz	CMR ² DC	CMR ² Linefreq	CMR ² 400 Hz
1	20 000 000	80	50	140	160	110
2	2 000 000	70	40	140	150	100
3	200 000	60	30	140	140	90
4	20 000	-	-	140	80	60
5	20 000	-	-	140	80	60
6	2 000	-	-	140	80	60

Transfer accuracy at Speed 1. Within 10 min and at $T_{ref} \pm 0.5^\circ C$ and within 10% of reference value

Range	ppm of reading + ppm of range
20 mV	20 + 5
200 mV	10 + 2
2 V	5 + 2
20 V	5 + 2
200 V	8 + 2
2000 V ²	8 + 2

Maximum Input Voltage⁴

	Rated, Input
V and V_{com}	1000 V _{peak}
V and ground	1000 V _{peak}
V _{com} and ground	450 V _{peak}
V _{com} and guard	60 V _{peak}
guard and ground	450 V _{peak}

Reading rates

Speed	Mode	Aperture	Max Scale length	Readings/sec
1	Direct/Burst	100 x 20 ms	20 000 000	0.2
2	Direct/Burst	10 x 20 ms	2 000 000	2
3	Direct/Burst	20 ms	200 000	20
4	Direct/Burst	2 ms	20 000	200
5	Direct/Direct	200 µs	20 000	1 000
6	Burst	200 µs	20 000	4 000
6	Direct	50 µs	2000	1 000
6	Burst	50 µs	2000	50 000

Response time (When stugl trigger signal and step-input are coincidental and analog filter off)

Speed	Excluding	Ranging	Including	Ranging	Digits from final value
	Internal trig.	Single trig.	Internal trig.	Single trig.	
1	5.5 s	5 s	6 s	5.5 s	50
2	1 s	0.5 s	1.5 s	1 s	10
3	110 ms	60 ms	600 ms	550 ms	10
4	15 ms	10 ms	500 ms	500 ms	5
5	3 ms	2 ms	500 ms	500 ms	5
6	1 ms	1 ms	500 ms	500 ms	5

Resistance

Two wire and Four-wire Ohms

Range	Full scale	Maximum resolution	Current Source	Temperature coefficient ¹
20 Ω	2 000 000	10 µΩ	10 mA	20
200 Ω	2 000 000	100 µΩ	1 mA	10
2 kΩ	20 000 000	100 µΩ	1 mA	10
20 kΩ	20 000 000	1 mΩ	100 µA	10
200 kΩ	20 000 000	10 mΩ	10 µA	10
2 MΩ	20 000 000	100 mΩ	1 µA	30
20 MΩ	20 000 000	1 Ω	100 nA	30
200 MΩ	200 000	1 kΩ	10 nA	100

General

Ranging: Manual on all ranges. Fast Automatic on 200 mV - 1000 V. (In Speed 1 and for 200 mV range manual down ranging only).
Max. NM signal: Peak value 140% of range
Nulling: Offset voltage and thermal voltage can be compensated in all ranges.
Filters: Selectable Analog filter. Above 5 Hz 60 dB/decade
Overload indication: If signal is too large measurement is performed but indicated with "OL".

¹ Measured with ±0.1% of selected line frequency or ±1% of 400 Hz
² Maximum input voltage 1000V
³ Measured with 1 kΩ unbalance and guard connected to CM voltage. Maximum Slew Rate 10 KV/µs
⁴ Max. allowable V-Hz product 10⁶. Max. rate of input voltage 5kV/µs.
⁵ Additional error from T_{ref} ±5°C

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Reading rates

Speed	Mode	Aperture	Max Scale length	Readings/sec
1	Direct	100 x 20 ms	20 000 000	0.2
2	Direct	10 x 20 ms	2 000 000	2
3	Direct	20 ms	200 000	20
4	Direct	2 ms	20 000	200
5	Direct	200 µs	20 000	1 000

Response time (When stugls trigger signal and step-input are coincidental for 200 Ω upto 2000 Ω)

Speed	Excluding Ranging		Including Ranging		Digits from final value
	Internal trig.	Single trig.	Internal trig.	Single trig.	
1	5.5 s	5 s	6 s	5.5 s	50
2	1 s	0.5 s	1.5 s	1 s	10
3	110 ms	60 ms	600 ms	550 ms	10
4	15 ms	10 ms	500 ms	500 ms	5
5	3 ms	2 ms	500 ms	500 ms	5

General

Ranging: Manual or Fast Automatic on all ranges.

Nulling: Offset voltage and thermal voltage can be compensated in all ranges.

Filters: Selectable Analog filter. Above 5 Hz 60 dB/decade.

Overload indication: If signal is too large measurement is performed but indicated with "OL".
True Ohm facility: Via "NULL" function.
Open circuit voltage: < 10 V

Maximum allowable lead resistance in 4-wire configuration: 100 Ω (10 Ω on 20 Ω range)
Protected: up to 500 Vac or dc, or 750 V_{peak}

DCCurrent

General

Range	Full scale	Maximum resolution	Burden voltage	Temperature coefficient ¹
2 µA	2 000 000	1 pA	< 0.25 mV	30 ± 5
20 µA	20 000 000	1 pA	< 2.5 mV	30 ± 5
200 µA	20 000 000	10 pA	< 25 mV	30 ± 5
2 mA	20 000 000	100 pA	< 250 mV	30 ± 5
20 mA	20 000 000	1 nA	< 250 mV	30 ± 5
200 mA	20 000 000	10 nA	< 50 mV	30 ± 5
2 A	20 000 000	100 nA	< 600 mV	30 ± 5

Reading rates

Speed	Mode	Aperture	Max Scale length	Readings/sec
1	Direct	100 x 20 ms	20 000 000	0.2
2	Direct	10 x 20 ms	2 000 000	2
3	Direct	20 ms	200 000	20
4	Direct	2 ms	20 000	200
5	Direct	200 µs	20 000	1 000

Maximum Input Voltage⁵

	Rated, Input
A and A _{com}	2.5 A _{peak}
A and ground	450 V _{peak}
A and guard	450 V _{peak}
A _{com} and ground	450 V _{peak}
A _{com} and guard	60 V _{peak}
guard and ground	450 V _{peak}

Accuracy¹ (ppm of Reading + ppm of Range).

Range	24 Hour	90 Day	1 Year
20 Ω	24 ± 5	36 ± 10	85 ± 10
200 Ω	24 ± 5	36 ± 10	85 ± 10
2 kΩ	24 ± 5	36 ± 10	85 ± 10
20 kΩ	24 ± 5	36 ± 10	85 ± 10
200 kΩ	24 ± 5	36 ± 10	85 ± 10
2 MΩ	72 ± 5	120 ± 10	240 ± 10
20 MΩ	72 ± 5	120 ± 10	240 ± 10
200 MΩ	360 ± 100	1000 ± 100	1000 ± 100

Maximum Input Voltage

	Rated, Input
Ω and Ω _{com}	750 V _{peak}
Ω and ground	1000 V _{peak}
Ω and guard	1000 V _{peak}
Ω _{com} and ground	450 V _{peak}
Ω _{com} and guard	60 V _{peak}
guard and ground	450 V _{peak}

Accuracy² (ppm of Reading + ppm of Range).

Range	24 Hour	90 Day	1 Year
2 µA	20 ± 10	100 ± 10	150 ± 10
20 µA	20 ± 10	100 ± 10	150 ± 10
200 µA	20 ± 10	100 ± 10	150 ± 10
2 mA	20 ± 10	100 ± 10	150 ± 10
20 mA	30 ± 30	150 ± 30	250 ± 30
200 mA	30 ± 30	150 ± 30	250 ± 30
2 A	30 ± 30	150 ± 30	250 ± 30

Noise rejection⁴ (dB)

Speed	Max. scale length	NMR Line freq.	NMR 400 Hz
1	20 000 000	80	50
2	2 000 000	70	40
3	200 000	60	30
4	20 000	-	-
5	20 000	-	-
6	2 000	-	-

¹ Specifications are for speed 1 and 2 and relative to calibration values.

² Additional error from T_{ref} ±5°C

³ Specifications are for speed 1,2 and 3

⁴ Measured with ±0.1% of selected line frequency or ±1% of 400 Hz

⁵ A-input protected with fuse 2.5 A/F

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Response time (When single trigger signal and step-input are coincidental)

Speed	Excluding Ranging		Including Ranging		Digits from final value
	Internal trig.	Single trig.	Internal trig.	Single trig.	
1	5.5 s	5 s	6 s	5.5 s	50
2	1 s	0.5 s	1.5 s	1 s	10
3	110 ms	60 ms	600 ms	550 ms	10
4	15 ms	10 ms	500 ms	500 ms	5
5	3 ms	2 ms	50 ms	500 ms	5

True-RMS AC Voltage (dc or ac Coupled)

General

Range	Full scale	Maximum resolution	Temperature coefficient ¹	input impedance
20 mV	200 000	100 nV	±(50 + 50)	1 MΩ // 150 pF
200 mV	2 000 000	100 nV	±(50 + 50)	1 MΩ // 150 pF
2 V	2 000 000	1 µV	±(50 + 50)	1 MΩ // 150 pF
20 V	2 000 000	10 µV	±(50 + 50)	1 MΩ // 150 pF

General

Ranging: Manual on all ranges. Fast Automatic on 2 µA - 2 A. (In Speed 1 and for 2 µA range manual down ranging only)

Max. NM signal: Peak value 140% of range

Overload indication: If signal is too large measurement is performed but indicated with "OL"

200 V	2 000 000	100 µV	±(50 + 50)	1 MΩ // 150 pF
2000 V ¹	2 000 000	1 mV	±(50 + 50)	1 MΩ // 150 pF

Accuracy³ (% of Reading + % of Range)

Range	24Hour	90days	1Year	
Fast response	400 Hz - 20 kHz	0.05 + 0.01	0.07 + 0.02	0.1 + 0.03
Normal response	40 Hz - 20 kHz	0.05 + 0.01	0.07 + 0.02	0.1 + 0.03
Slow response	10 Hz - 40 Hz	0.07 + 0.01	0.1 + 0.02	0.15 + 0.03
	40 Hz - 20 kHz	0.05 + 0.01	0.07 + 0.02	0.1 + 0.03
Fast, Normal and slow	20 kHz - 50 kHz	0.125 + 0.025	0.175 + 0.05	0.25 + 0.075
	50 kHz - 100 kHz	0.25 + 0.05	0.35 + 0.1	0.5 + 0.15
	100 kHz - 300 kHz	0.75 + 0.15	1.05 + 0.3	1.5 + 0.45
	300 kHz - 500 kHz	1.25 + 0.25	1.75 + 0.5	2.5 + 0.75
	500 kHz - 1 MHz	2.5 + 1	3.5 + 2	5 + 3

Reading rates

Speed	Mode	Aperture	Max Scale length	Readings / Sec
1	Direct	100 x 20 ms	2 000 000	0.2
2	Direct	10 x 20 ms	200 000	2
3	Direct	20 ms	20 000	20

AC Current

General⁶

Range	Full scale	Maximum resolution	Burden voltage	Temperature coefficient ¹ (% of reading)/K
200 µA	2 000 000	100 pA	< 25 mV	0.01%
2 mA	2 000 000	1 nA	< 250 mV	0.01%
20 mA	2 000 000	10 nA	< 250 mV	0.01%
200 mA	2 000 000	100 nA	< 350 mV	0.01%
2 A	2 000 000	1 µA	< 600 mV	0.01%

Maximum Input Voltage⁴

	Rated, Input
V and V _{com}	1000 V _{peak}
V and ground	1000 V _{peak}
V and guard	1000 V _{peak}
V _{com} and ground	450 V _{peak}
V _{com} and guard	60 V _{peak}
guard and ground	450 V _{peak}

General

Crest Factor: > 4 (At full scale. Increasing to down scale via 4 x full scale/reading). Warning on display when clipping occurs.

Common Mode Rejection: At dc > 120 dB, at line frequency > 60 dB.

Filter: Selectable for low frequency limits of 10 Hz, 40 Hz or 400 Hz.

Ranging: Manual and fast automatic on all ranges (in Speed 1 and for 20 mV range manual ranging only).

¹ Additional error from T_{ref} ±5°C for frequency range up to 20 KHz

² Maximum input voltage 700V.

³ For 20 mV range add 0.01% of range. For DC coupled measurements, add ±0.05% of reading +0.01% of range for DC components.

⁴ Max. allowable V-Hz products 2 x 10⁷. Max. slew of input 5 kV/µs.

⁵ Specifications are for speed 1 (0.2 meas/sec.)

⁶ Additional error from T_{ref} ±5°C

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Accuracy (% of Reading + % of Range)

Range		24 Hour	90days	1 Year
Fast response	400 Hz - 1 kHz	0.07 + 0.05	0.1 + 0.05	0.15 + 0.05
Normal response	40 Hz - 1 kHz	0.07 + 0.05	0.1 + 0.05	0.15 + 0.05
Slow response	10 Hz - 40 Hz	0.1 + 0.05	0.15 + 0.05	0.2 + 0.05
	40 Hz - 1 kHz	0.07 + 0.05	0.1 + 0.05	0.15 + 0.05
Fast, Normal	1 kHz - 5 kHz	0.1 + 0.1	0.1 + 0.1	0.1 + 0.1
slow	5 kHz - 10 kHz	0.45 + 0.1	0.5 + 0.2	0.5 + 0.2

Reading rates

Speed	Mode	Aperture	Max Scale length	Readings / sec
1	Direct	100 x 20 ms	2 000 000	0.2
2	Direct	10 x 20 ms	200 000	2
3	Direct	20 ms	20 000	20

General

Crest Factor: > 3.3 (At full scale. Increasing to down scale via 3.3 x full scale/reading). Warning on display when clipping occurs.

Common Mode Rejection: At dc > 120 dB, at line frequency > 60 dB.

Filter: Selectable for low frequency limits of 10 Hz, 40 Hz or 400 Hz.

Ranging: Manual and fast automatic on all ranges.

Very Low Frequency Voltage (VLF).

General

Range	Full scale	Maximum resolution	Temperature coefficient (ppm/reading + ppm/range)/K	input impedance ¹
100 mV	1000	10 µV	±(20 + 10)	1 GΩ // 150 pF
1 V	10 000	100 µV	±(20 + 10)	1 GΩ // 150 pF
10 V	10 000	1 mV	±(20 + 10)	1 GΩ // 150 pF
100 V	10 000	10 mV	±(20 + 10)	10 MΩ // 150 pF
1000 V	10 000	100 mV	±(20 + 10)	10 MΩ // 150 pF

Accuracy (% of Reading + % of Range)

Range	24 Hour	Speed 1 and 2	90 Day	1 Year
100 mV	0.08 + 0.02	0.1 + 0.03	0.15 + 0.05	0.15 + 0.05
1 V	0.08 + 0.02	0.1 + 0.03	0.15 + 0.05	0.15 + 0.05
10 V	0.08 + 0.02	0.1 + 0.03	0.15 + 0.05	0.15 + 0.05
100 V	0.08 + 0.02	0.1 + 0.03	0.15 + 0.05	0.15 + 0.05
1000 V	0.08 + 0.02	0.1 + 0.03	0.15 + 0.05	0.15 + 0.05

Frequency Range: 2 mHz ... 500 Hz.

Common Mode Rejection: At dc > 140 dB, at selected line frequency > 80 dB.

Filter: Selectable Analog filter. Above 5 Hz 60 dB/decade.

Ranging: Manual and fast automatic on all ranges.

Overload indication: If signal is too large measurement is performed but indicated with "OL."

Frequency/Period.

General

Speed	Meas/sec	Range	Maximum Resolution
1	1..2	2 Hz ... 2 MHz	1 µHz
2	10..20	20 Hz ... 2 MHz	100 µHz
3	100..200	200 Hz ... 2 MHz	10 mHz
1	1..2	500 ns ... 0.5 s	0.1 ps
2	10..20	500 ns ... 5 ms	1 ps
3	100..200	500 ns ... 5 ms	10 ps

Input characteristics: The frequency or period of the measured signal is derived from an AC voltage measurement. Zero-crossings of the signal are used to determine the measurement result.

Voltage sensitivity: 2% of fullscale.

Accuracy: 10 ppm + 1 count.

Ranging: Automatic on all ranges.

¹ Additional error from T_{ref} ±5°C

² Max. allowable V-Hz products 10⁶. Max. slew of input 5 kV/µs.

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Temperature

Measuring principle: Pt100 (PRT).

Range: -20°C ... +850°C

Resolution: 0.1°C or 0.01°C

Full Scale: 850.0°C

Accuracy: ±(0.03% of reading + 0.02°C excl. probe)

Temperature Coefficient: ±(0.001% of Reading + 0.001% of Range)/K

Measuring Current: 1 mA

Measuring speed: 2 or 20 measurements/sec

Linearisation: According DIN 43760, NBS390P or

Where "X" is the measured value. This makes it possible to scale transducers output to display in engineering units.

Delta %: Difference of measured value will be given in percentage of a measured or programmed value.

dBm: Measured value will be given as dBm where 0 dBm is 1 mW in 600 Ω reference. Reference resistance can be changed via keyboard or interface. This measurement can be combined with the „Relative“ function giving a relation to a measured or programmed value.

Relative ("ZERO"): Measured or programmed value can be used as a ZERO-reference for new

Ordering Information

Models

PM 2530/... System Digital Multimeter.

Power Options

Instruments are normally delivered in accordance with local power requirements. If alternative power option is required, the instrument can be supplied with different power options.

NBS392P1.
Measurement Presentation: °C, °F or K.
Max Voltage (Imp/guard): 60 V_{peak}

General

Filter options: Selectable in all functions:

Digital: Weighted ($\leq X \leq N$): $\frac{w-1}{w} \cdot \leq X_{n-1} + \frac{1}{w} \cdot X_n$)
w is programmable between 1 and 100.
 $X_1 + X_2, \dots, + X_k$
or: Average $\langle y \rangle = \frac{X_1 + \dots + X_k}{k}$

k is programmable between 1 and 100.

Min / Max: Selectable for all functions. Constant recording of highest and lowest measured value since last function selection or reset.

Triggering: Internal, External, via interface, via BNC connector or Manual. Start of measurement cycle can be synchronized with trigger signal.

Relative measurements: For all functions, a reference value can be used as "ZERO". Value can be a measured value or a manually or via IEEE-488 bus given value.

Timing: Delay after trigger and Measurement interval can be programmed.

Burst mode: In the burst mode up to 15000 measurement can be made and stored in a buffer with programmable delay and programmable interval between measurements. Number of measurements after a trigger signal can be programmed.

Limits: Limits can be set. "Exceed limit" warning available at display and/or external connector. Total number of measurement, number of "passed", "Exceed high" and "Exceed low" is stored.

Post Processing:

Scale factor can be programmed. The measured value can be processed with the formula: $\frac{AX \pm B}{Z}$

measurements.
Save and Recall: Up to 10 settings of the instrument (including relevant auxiliary functions and parameters) can be stored of which 2 in a non volatile memory and recalled for later use. This makes repeating of the same measurement easier and makes it possible to communicate via the interface very quickly. Settings are stored under the numbers 0 ... 9. The setting stored under "0" will be used as default at power up.

Operating Environment.

Reference temperature: 23 °C $\pm$ 1 °C

Rated range of use: 0 °C ... 50 °C

Limit range of operation: 0 °C ... 55 °C

Limit range of storage and transport: -40 °C ... + 70 °C

Operating Humidity: range up to 75% at 40 °C

Warm-Up time: 1 Hour to reach 90 days accuracy

Safety: Meets requirements of IEC 348, VDE 0411 and UL 1244. Approved by CSA according to CSA 22.2 Nr 231.

Standards: Meets requirements of MIL-T-28800D type III, class 5, Style E, Color R.

Altitude (MIL-T-28800D): 7.3

Operating: 3050 m (10000 ft)

Non operating: 4570 m (15000 ft)

EMI

Emission: VDE 0871 class B + CISPR 11

Susceptibility: IEC 8011-6 exc part 2

Electrostatic discharge: IEC 801-2

General Specifications

Power requirements: 103...132 V, or 195-258 V ac, 50, 60 or 400 Hz.

Size (h x w x d): 8.6 x 42 x 31 cm 3.4 x 16.5 x 11.4 in.

Weight: 5 kg (11 lbs)

IEEE-488 Interface Function Complies with: IEEE-488.2 (1987) SCPI (1990) compatible Allows complete control and data output capability, and supports the following interface function subsets: SH1, AH1, T6, L4, SR1, RL1, DC1, DT1, PP0, E2.

measurements.
please specify one of the following options:
/011 Universal European 230V/16A/50 Hz
/013 Standard North America 115V/16A/60 Hz
/014 United Kingdom 230V/13A/50 Hz
/015 Switzerland 230V/10A/50 Hz.

Accessories

Delivered with the instrument

PM 9266/03: Set of UL silicon isolated safety test leads. Pair of detachable probes and adaptors to "non safety" 4 mm banana plugs are supplied with the leads.

Operating manuals in English, French and German. Quick operating guides in English, French and German.

4822 070 12501 Spare fuse 2.5 AF

4822 070 31601 Spare fuse 160 mA T (For 230 V)

4822 070 36309 Spare fuse 315 mA T (For 120 V)

Optional available

PM 9071 Measuring cable. Banana (m) - banana (m) 135 Ω

PM 9072 Measuring cable. Banana (m) - BNC (m) 135 Ω

5440A-7002 Low thermal Cable Set.

PM 9268 4-Wire test lead with 4-mm banana plugs for resistance measurements with Kelvin clips.
PM 9252 Pt100 temperature probe with 4-mm banana plugs.

Y9109 Adapter banana (m) - BNC (f)

Y8133 Test Lead Kit. With leads with standard banana plugs at one end and safety banana plugs on the other end. Probes, alligator clips and a spring loaded hook clip.

Y8140 Test Lead Set. Slim line with needle pins.

83 RF High frequency probe. 100 kHz ... 10 MHz.

85 RF High frequency probe. 100 kHz ... 500 MHz.

80T-1500 Temperature probe °C, °F or off, -50°C ... 150°C

Y8100 DC or AC current probe, DC, 400 Hz, 200 A max.

Y8101 AC current probe 48 ... 1200 Hz, 150 A max.

80I-600 AC current probe. 30 ... 10.000 Hz, 600 A max.

4822 872 30448 Operating Manual English

4822 872 30449 Quick Operating Guide English

4822 872 30451 Operating Manual German

4822 872 30452 Quick Operating Guide German

4822 872 30453 Operating Manual French

4822 872 30454 Quick Operating Guide French

4822 872 35315 Service Manual English

¹⁾ Storage for LCD display -25% ... + 70 °C

Bench Multimeters

Author: Peter J. Vis

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